

The Economics of Expertise in the Age of Generative AI

From Constrained Scientific Creativity to Epistemic Scarcity

Yaoharee Lahtee Independent Scholar, Bangkok, Thailand; Open Civil Science Initiative
ORCID: 0009-0005-3861-0626 **Manuscript status:** Final Journal Manuscript v1.1. This is a submission-ready theoretical manuscript, not a claim of peer-reviewed acceptance. Existing equations attributed to the Human–AI Readout Programme are provenance-checked against Toledo. Equations introduced in this manuscript are explicitly labelled as new definitions, derivations, propositions, or empirical hypotheses.

Version note (v1.1: Uplift edition — changes from v1.0): (1) added a provenance footnote at Equation (6) clarifying that the cited knowledge triple records an earlier formulation of Toledo code `weld/E.03.v1`, whose current canonical statement has since been superseded by a richer tuple for the same admitted object; (2) added a DOI or ISBN identifier to the four references that previously carried neither (Arrow, 1962; Collins & Evans, 2007; Polanyi, 1966; Stokes, 1997); (3) added an in-text citation to Edwards and Roy (2017) in Section 1, resolving that reference's prior orphan status; (4) added, to the Appendix A ledger's Provenance column, the Toledo codes now assigned to this paper's own equations (`weld/H.22.v1` – `weld/H.29.v1`, `weld/W.03.v1` – `weld/W.10.v1`) and to its Core Epistemic Registration object (`weld/H.30.v1` – `weld/H.34.v1`); (5) updated the Toledo citation to the current release, Version 1.4.0, Zenodo version DOI 10.5281/zenodo.22637913. No other sentence, equation, or reference in this manuscript was altered; all v1.0 content not listed above is unchanged.

Core Epistemic Registration

This manuscript uses the three-part project registration proposed in the paper itself. Registration is an attribution device, not a certification of truth or expertise.

Core Respondent / Experience-Based Expert: Yaoharee Lahtee — self-registered practice anchor based on direct experience in independent AI-assisted knowledge production, practice-linked inquiry, and the organization of research outside conventional university infrastructure. This registration identifies the source of the paper's experiential problem framing; it does not by itself validate any theoretical claim.

Interactional Expert: None

AI Model(s) Used: GPT-5.6 Sol — conceptual synthesis, equation-provenance audit, rival-model generation, mathematical restructuring, literature organization, adversarial review, and manuscript drafting assistance.

AI Model(s) Used (continued): Claude Fable 5.1 (via Claude Code) — independent glosa review (rules 6/9/17, tier and Toledo-provenance checks), registration of this paper's

equations in Toledo, PDF typesetting, and Zenodo deposit

Non-collapse rule:

Core experiential source $\neq$ Interactional expert $\neq$ AI model.

Abstract

Contemporary academic systems can make strategic conformity a rational response for early-career researchers by rewarding short-cycle, tractable, easily evaluated outputs while imposing career, funding, publication, and hierarchical risks on high-variance inquiry. Haubrock and colleagues recently synthesize this structural problem and call for more stable career pathways, decentralized decision-making, and evaluation that better recognizes originality, persistence, collaboration, and nonlinear scientific careers. This paper takes that diagnosis as an anchor and asks what changes when generative artificial intelligence sharply lowers the cost of producing plausible answers, syntheses, code, and candidate hypotheses.

I develop an **economics of expertise** in which AI-generated knowledge-like output is formally separated from validated knowledge, and knowledge is separated from expertise. The central mechanism is an **AI-induced epistemic scarcity shift**. Let Λ_t denote effective candidate-validation workload generated per period and μ_t the system's capacity to discriminate, test, correct, and ground candidate claims. A candidate backlog evolves as

$$B_{t+1} = \max\{0, (1 - \delta_B)B_t + \Lambda_t - \mu_t\}.$$

In the positive-backlog steady state, an AI capability shock z increases unresolved epistemic workload when

$$\frac{\partial \Lambda}{\partial z} > \frac{\partial \mu}{\partial z}.$$

The theory therefore does not assume that AI necessarily worsens knowledge quality. It specifies the condition under which AI relocates scarcity from answer generation toward validation, judgment, world contact, correction, and contributory expertise. Collins and Evans's distinction among no expertise, interactional expertise, and contributory expertise supplies the human expertise layer, while a mandatory three-part Core Epistemic Registration makes visible the experiential problem source, the presence or absence of a human interactional bridge, and every materially used AI model.

For practice-linked research, the resulting architecture is **Live Problem** $\rightarrow$ **Practitioner + Researcher** $\rightarrow$ **AI-assisted Inquiry** $\rightarrow$ **World Test** $\rightarrow$ **Feedback** $\rightarrow$ **Revision**, with a separately protected open-exploration track for basic research. The framework generates falsifiable propositions, comparative statics, a multi-university empirical programme, and coordinated reforms for researchers and practitioners, universities, and public funders. The central claim is that generative AI does not abolish the economics of knowledge. It changes the location of its bottleneck, and universities designed around the old scarcity

regime risk becoming more efficient at producing outputs without becoming more effective at producing world-grounded expertise.

Highlights

- Extends the structural-creativity problem of early-career researchers into an economics of expertise for the generative-AI era.
 - Separates AI-generated candidate abundance from validated knowledge and from human expertise.
 - Derives a candidate-validation backlog and a condition under which AI increases epistemic scarcity rather than reducing it.
 - Integrates interactional and contributory expertise into the economics of validation without treating credentials as expertise.
 - Proposes a three-part Core Epistemic Registration: experiential problem source, interactional expert or `None`, and all materially used AI models.
 - Replaces publication-first architecture in practice-linked research with live-problem coupling, world testing, and revision, while protecting basic research from an impact-or-perish regime.
-

1. Introduction

Scientific institutions do not merely allocate money. They allocate time, risk, attention, legitimacy, access to equipment, access to collaborators, and permission to fail. These allocations shape which questions researchers can afford to ask. For early-career researchers, the effect is particularly consequential because the period in which scientific independence is supposed to emerge often coincides with the period of greatest employment insecurity, dependence on supervisors, grant uncertainty, and pressure to accumulate visible outputs.

Haubrock et al. (2026) frame this as a systems problem. Short funding cycles, productivity-oriented evaluation, risk-averse review, publication costs, institutional inequality, precarious employment, and hierarchical dependence can jointly make strategic conformity rational, consistent with the climate of perverse incentives and hypercompetition documented across contemporary academic research systems (Edwards & Roy, 2017). Under such conditions, scientific creativity is not simply a trait that some individuals possess and others lack. It is an emergent outcome of an institutional environment that changes the relative price of intellectual risk.

This paper accepts that diagnosis as its anchor. It argues, however, that generative AI changes the underlying economic problem. Academic systems were built in a period when searching literature, translating across fields, producing polished prose, writing code, synthesizing sources, and generating plausible candidate explanations were costly

activities. Generative AI lowers many of those costs. Evidence from professional writing and customer-support settings already demonstrates large productivity effects, often with heterogeneous gains across skill levels (Noy & Zhang, 2023; Brynjolfsson, Li, & Raymond, 2025). Generative AI can also increase individual creative performance while reducing diversity at the collective level (Doshi & Hauser, 2024).

The relevant policy question is therefore no longer only whether AI helps researchers produce more. It is:

Which epistemic activities become abundant, which remain scarce, and which institutional rewards become misaligned when plausible candidate production becomes cheap?

The answer developed here is an **AI-induced epistemic scarcity shift**. AI can make candidate production abundant without making validation, world testing, correction, or expertise equally abundant. If candidate workload expands faster than validation capacity, the bottleneck moves downstream. In that regime, the scarce resources are no longer mainly the capacity to generate academically plausible answers. They are the capacities to discriminate among them, identify decisive evidence, connect them to world constraints, revise after failure, and assume responsibility for action.

This argument connects five literatures that are usually discussed separately. The economics of science explains why knowledge production depends on public goods, incentives, priority systems, career structures, and institutional design (Nelson, 1959; Arrow, 1962; Dasgupta & David, 1994; Stephan, 2012). The economics of innovation shows why tolerance for early failure and long horizons can support exploration (Manso, 2011; Azoulay, Graff Zivin, & Manso, 2011). Studies of expertise and experience distinguish between the ability to speak a domain's language and the ability to contribute to its practice (Collins & Evans, 2002, 2007). Organizational learning and tacit-knowledge traditions show why external information is not automatically converted into usable capability (Polanyi, 1966; Cohen & Levinthal, 1990). Research-policy literatures show that institutional bridges can reorganize knowledge production across universities, firms, government, and use-inspired research settings (Stokes, 1997; Etzkowitz & Leydesdorff, 2000; Bikard, Vakili, & Teodoridis, 2019).

The missing link is a model of what happens when AI changes the cost of **candidate epistemic production** faster than it changes the cost of **validation and expertise formation**.

The paper makes four contributions.

First, it formalizes the non-collapse:

$$K_{\text{like}}^{\text{AI}} \neq K_{\text{validated}}.$$

Second, it introduces a workload-based candidate-validation model that yields explicit comparative statics. This avoids the strong and unnecessary assumption that all candidate

claims are homogeneous.

Third, it embeds expertise into the model without reducing expertise to a scalar stock of information. The human state is indexed by question, domain, and time, and interactional expertise is distinguished from contributory expertise.

Fourth, it translates the theory into research-system design. For practice-linked research, the sequence

Researcher → Topic → Funding → Research → Paper → Find User

should cease to be the default architecture. The alternative is

Live Problem → Practitioner + Researcher → AI-assisted Inquiry → World Test → Feedback →
--

This is not a claim that all science should be applied. The framework explicitly protects open exploration and basic research. The argument is narrower: when research is intended to address a consequential real-world problem, world contact should enter at the beginning and remain active throughout the epistemic cycle rather than appear only after publication.

2. From Constrained Creativity to Constrained Expertise Formation

2.1 The anchor problem

Haubrock et al. (2026) assemble a broad structural diagnosis of why contemporary academic systems can constrain scientific creativity and hold back early-career researchers. The central mechanisms include short funding horizons, output-oriented evaluation, publication pressure and cost, hierarchical dependence, career precarity, risk-averse review, unequal institutional resources, and concentration of talent in already advantaged settings.

A crucial insight is that conformity can be individually rational. A researcher facing a short contract does not choose between "creative" and "uncreative" projects in an abstract intellectual vacuum. The choice is conditioned by completion time, data access, reviewer familiarity, supervisor support, publication probability, funding eligibility, and the probability that failure will occur before the next career gate.

The result is an institutional selection process. High-risk research may have high social value but low private career value to the person who must choose it.

This is consistent with established economic evidence. Manso (2011) shows theoretically that innovation incentives require tolerance for early failure and reward for long-term success. Azoulay et al. (2011), comparing funding environments in the life sciences, find that greater freedom and tolerance of early failure can induce more exploratory research.

Wang, Veugelers, and Stephan (2017) show that highly novel work has higher upside but also greater variance, delayed recognition, and disadvantages under short-window bibliometric evaluation. Aghion, Dewatripont, and Stein (2008) model academic freedom as a control-rights arrangement that can protect early-stage exploration from excessive direction.

The structural-creativity problem therefore already has an economic core: institutions change the payoff distribution associated with exploration.

2.2 The deeper mechanism: selection before explicit choice

The relevant set of research questions is not the set of all scientifically possible questions. It is the subset that remains feasible, legible, fundable, and career-survivable for a particular researcher at a particular time.

A question may be physically researchable but effectively absent because it requires equipment the researcher cannot access, a practitioner network the university does not possess, a time horizon longer than the grant, an unfamiliar method that reviewers distrust, or a cross-disciplinary vocabulary no local mentor can translate.

Institutional power can therefore act before explicit choice. Repeated incentives alter the distribution of live research possibilities.

This shifts the problem from creativity alone to **expertise formation**. If early-career researchers repeatedly optimize for low-variance, output-legible tasks, they may accumulate literature familiarity and publication skill without accumulating equivalent capacity for world collision, correction, transfer, and cross-domain discrimination.

The key question becomes:

What kind of researcher does the incentive system train a person to become?

2.3 Why generative AI changes the mechanism

Generative AI lowers the cost of activities that academic systems already reward: searching, drafting, coding, summarizing, translating, generating candidate explanations, and formatting outputs. If reward functions remain unchanged, measured productivity can increase without a proportional increase in validated knowledge or expertise.

This creates a new risk. The university can become increasingly efficient at producing the visible signals that its own evaluation system counts while the scarce activities required to ground those signals remain underfunded.

The danger is not simply AI hallucination. It is **institutional amplification of an obsolete production function**.

3. Positioning the Theory in the Global Literature

3.1 Economics of knowledge and science

The classic economics of science begins from the unusual properties of information and scientific knowledge. Nelson (1959) and Arrow (1962) explain why private incentives can underprovide basic research when knowledge is difficult to appropriate and produces spillovers. Dasgupta and David (1994) extend this into an institutional economics of open science in which priority, disclosure, and reputational rewards help sustain scientific production. Stephan (2012) brings the analysis to contemporary career systems, showing that scientists and institutions respond to monetary and reputational incentives, employment structures, grant competition, and risk.

This literature establishes that the organization of knowledge production matters. It does not, however, assume a world in which machines can generate large volumes of plausible epistemic objects at near-zero marginal cost relative to historical research workflows.

The present theory therefore does not replace the economics of science. It adds a new scarcity distinction inside it:

candidate production capacity $\neq$ validation capacity.

3.2 Burden of knowledge, absorptive capacity, and tacit practice

Jones (2009) argues that the growing burden of accumulated knowledge pushes innovators toward longer training, specialization, and teamwork. Cohen and Levinthal (1990) show that the ability to recognize, assimilate, and apply external information depends on prior related knowledge: access to information is not equivalent to absorptive capacity. Polanyi's account of tacit knowledge similarly warns against identifying articulated information with the whole of practical knowing.

Generative AI intensifies the importance of these distinctions. AI may reduce the cost of accessing and articulating explicit knowledge while leaving important components of practical capability dependent on practice, embodiment, judgment, institutional permission, and repeated correction.

3.3 Expertise and experience

Collins and Evans's Studies of Expertise and Experience programme provides a vocabulary for separating linguistic-cultural competence from practical contribution. **Interactional expertise** is the ability to master the language and conceptual practices of a domain sufficiently to engage meaningfully with contributory experts. **Contributory expertise** includes the practical competence required to participate in and contribute to the domain's work (Collins & Evans, 2002, 2007).

This distinction is critical for AI. A model may generate language that resembles highly fluent interactional performance, but linguistic performance by a model does not thereby

establish human interactional expertise, contributory expertise, or validated domain judgment. Likewise, a credential does not automatically establish expertise outside the domain in which it was earned.

The present paper uses the Collins–Evans categories as ideal-type research readouts, not as naturally discrete ontological states.

3.4 Use-inspired research and institutional bridges

Stokes (1997) rejects a simple basic-versus-applied dichotomy and identifies a use-inspired basic-research space in which fundamental understanding and practical relevance can coexist. The Triple Helix literature highlights dynamic relations among universities, industry, and government (Etzkowitz & Leydesdorff, 2000). Empirical work on university–industry collaboration shows that cross-institutional partnerships can, under some conditions, support productive specialization rather than merely compromise open science (Bikard et al., 2019).

These literatures establish that practice linkage is not inherently anti-scientific. The unresolved issue is how to specify the **epistemic roles** inside such collaboration and how AI changes their relative costs.

3.5 Metrics, novelty, and scientific direction

Bibliometric evaluation can systematically favor short-term legibility. The Leiden Manifesto warns against allowing indicators to substitute for expert judgment and research missions (Hicks et al., 2015). Wang et al. (2017) show that novel science can be riskier, slower to be recognized, and disadvantaged by short citation windows. Park, Leahey, and Funk (2023) document a long-run decline in the disruptiveness of papers and patents and link this to narrowing use of prior knowledge.

These findings do not prove the present theory, but they establish a compatible empirical landscape: output volume can increase while the organization of search, novelty, and scientific direction deteriorates.

3.6 Generative AI and productivity

Noy and Zhang (2023) show substantial productivity gains from generative AI in professional writing tasks. Brynjolfsson et al. (2025) find significant productivity effects in customer support, with especially large gains among less-experienced workers. Doshi and Hauser (2024) find that generative AI can improve individual creativity while reducing collective diversity of content.

These studies demonstrate that AI can alter both the level and distribution of performance. They do not establish that immediate assisted performance becomes retained human capability or that candidate generation becomes validated knowledge.

The paper's central gap is therefore precise:

Existing literatures explain the economics of science, innovation incentives, expertise, absorptive capacity, university–industry relations, and AI productivity, but do not yet supply a common mechanism for the case in which AI expands candidate epistemic production faster than the system expands world-grounded validation capacity.

4. Provenance and Root-to-Domain Discipline

4.1 Equation-source rule

Existing Human–AI Readout Programme equations used in this paper are taken from **Toledo — Equation Library of the Human–AI Readout Programme** (Lahtee, 2026a). The repository's equation-source policy requires existing equations, symbols, constraints, and derivations to be verified there before use. New equations in this manuscript are labelled as such.

This rule matters because the paper distinguishes between:

- an existing equation or non-collapse;
- an economics-domain instantiation of an existing structural condition;
- a new definition introduced for this paper;
- a new derivation;
- and an empirical hypothesis.

4.2 Root-to-domain requirement

Economic objects such as cost, scarcity, expertise, funding, and validation are not treated as root ontological primitives. They are domain-level readouts.

Let S_t denote a retained state with admissible transition F_t , and let q_D be a declared domain translation. A domain reading should preserve the distinctions required by the claim. The generic commuting requirement is

$$q_{D,t+1} \circ F_t = F_{D,t}^{\#} \circ q_{D,t} \quad (1)$$

Status: economics-domain instantiation of the Toledo domain-weld requirement; not a new Toledo theorem.

The methodological implication is simple: an economic variable may summarize a deeper process, but it must not erase a distinction that later becomes causally or epistemically necessary.

4.3 Load-bearing non-collapses

The framework depends on four separations:

$$\boxed{K_{\text{like}}^{AI} \neq K_{\text{validated}}} \quad (2)$$

$$\boxed{\text{AI-assisted performance} \neq \text{retained human capability}} \quad (3)$$

$$\boxed{\text{credential} \neq \text{expertise}} \quad (4)$$

$$\boxed{\text{project role} \neq \text{expertise type} \neq \text{AI system}} \quad (5)$$

Equation (2) is an existing Toledo non-collapse. Equation (3) corresponds to the Toledo assisted-performance versus Human Return non-collapse. Equations (4)–(5) are methodological non-collapses motivated by the expertise literature and this paper's governance architecture.

5. Knowledge Is Not Expertise

A relevant Toledo knowledge representation is

$$\boxed{K_S^A = (Tr_A, Str_A, Cap_A)} \quad (6)$$

where Tr_A denotes retained trace, Str_A organized structure, and Cap_A capacity associated with the knowledge state.

Provenance footnote: this triple records an earlier formulation of the same admitted knowledge object under Toledo code `weld/E.03.v1` (alias `knowledge-admission`, historical lineage name `knowledge-triple`). Under Toledo's own "latest formulation wins" rule, that code's current canonical statement has since been superseded by a richer multi-field tuple for the same object. This paper cites the historical triple form deliberately, as the level of detail this manuscript's argument requires; a reader who looks up `weld/E.03.v1` today will find that richer current statement rather than (Tr_A, Str_A, Cap_A) , and should read this as lineage, not mismatch.

This paper deliberately does **not** define expertise as Equation (6). Doing so would collapse a knowledge architecture into a social-practical status without justification.

An earlier scalar candidate,

$$E = (Tr \cdot Str \cdot Cap)^{1/3},$$

is rejected as a foundational expertise equation because it assumes a common measurement scale, imposes substitutability, and destroys distinctions that may be causally important. Such a scalar could later be constructed as an empirical measurement readout, but only with declared normalization, domain, reader, thresholds, and validation evidence.

Expertise instead remains question-, domain-, and time-relative.

6. Expertise as a Context-Bound Human State

6.1 Three ideal-type states

For person i , question Q , domain D , and time t , define

$$\chi_i(Q, D, t) \in \{N, I, C\} \quad (7)$$

where:

- N = No Expertise;
- I = Interactional Expertise;
- C = Contributory Expertise.

Status: New Definition 1; a formal research readout of Collins–Evans ideal types.

The indexing is essential. The statement "person i is an expert" is incomplete without specifying the domain and task to which the claim refers.

6.2 Not a deterministic ladder

The sequence

$$N \rightarrow I \rightarrow C$$

is not treated as a universal developmental law.

Instead:

$$P(\chi_{i,t+1} = b \mid \chi_{i,t} = a, A_t, P_t, W_t, R_t) \quad (8)$$

where A_t denotes AI assistance, P_t practice exposure, W_t world feedback, and R_t relevant resources and institutional conditions.

Status: Open Proposition 1.

The expression permits progression, stagnation, regression, and domain-specific heterogeneity.

6.3 Interactional compression hypothesis

Generative AI can lower barriers to terminology, literature search, translation, explanation, repeated questioning, and comparison across sources. For some domains this should reduce the cost of reaching interactional competence.

$$C_{N \rightarrow I}^{AI} < C_{N \rightarrow I}^{baseline} \quad (9)$$

H1 — Interactional Compression. Equation (9) is an empirical hypothesis for declared tasks and domains, not a universal law.

6.4 Contributory persistence hypothesis

When contributory expertise depends strongly on embodiment, tacit coordination, equipment, responsibility, irreversible decisions, or repeated world collision, AI should reduce the cost of $I \rightarrow C$ less than it reduces $N \rightarrow I$:

$$|\Delta C_{I \rightarrow C}^{AI}| < |\Delta C_{N \rightarrow I}^{AI}| \quad (10)$$

H2 — Contributory Persistence. This is an open empirical hypothesis restricted to practice-intensive domains.

7. Core Epistemic Registration

7.1 Why project roles must be visible

Authorship alone does not show where direct problem experience entered a study, whether anyone could translate between the practitioner's world and the relevant disciplinary community, or which AI systems materially transformed the inquiry.

The paper therefore proposes a three-part registration for practice-linked and AI-assisted research.

Crucially, this is a **project-role registry**, not a fourth expertise taxonomy. A person may occupy multiple roles; the roles do not certify the truth of that person's claims.

7.2 Registration object

For project p :

$$\mathcal{R}_p = \langle E_p^{exp}, E_p^{int}, \mathcal{M}_p^{AI} \rangle \quad (11)$$

where:

- E_p^{exp} = Core Respondent / Experience-Based Expert, the registered source of direct problem experience;
- E_p^{int} = registered human Interactional Expert;
- $\mathcal{M}_p^{AI}$ = set of AI models materially involved.

If no human interactional expert is present:

$$E_p^{int} = \emptyset \quad (12)$$

If several models are used:

$$\mathcal{M}_p^{AI} = \{M_1, M_2, \dots, M_k\} \quad (13)$$

A generic statement such as "AI was used" is insufficient because model identity and role affect reproducibility, dependency, and provenance.

7.3 Registration does not confer authority

Registration answers three attribution questions:

1. Who brought direct experience of the problem?
2. Was a human interactional bridge present?
3. Which AI systems materially shaped the inquiry?

It does not answer whether the claims are true. An experience-based expert can overgeneralize. An interactional expert can misunderstand a contributory practice. Multiple models can converge on the same error. Registration makes the epistemic composition visible; validation remains separate.

8. A Formal Economics of Candidate Abundance and Validation Scarcity

8.1 Candidate sets and heterogeneous validation workload

A raw count of AI-generated claims is not an adequate economic variable because candidate claims differ in complexity, stakes, and validation cost.

Let $\mathcal{C}_t^{new}$ be the set of new candidate claims generated in period t . Each candidate c carries a validation workload $w(c) > 0$, expressed in declared validation-equivalent units.

Define effective candidate workload:

$$\Lambda_t = \sum_{c \in \mathcal{C}_t^{new}} w(c) \quad (14)$$

Status: New Definition 2.

The use of workload rather than claim count removes the assumption that all candidates are homogeneous.

8.2 Validation capacity

Define the system's validation capacity in the same workload units:

$$\mu_t = V(E_t^C, E_t^I, W_t, D_t, R_t, A_{v,t}) \quad (15)$$

where:

- E_t^C = available contributory expertise relevant to the claims;
- E_t^I = available interactional expertise;
- W_t = access to world-side testing;
- D_t = data and evidence access;
- R_t = non-AI resources such as time, equipment, money, permissions, and review attention;
- $A_{v,t}$ = AI capability allocated to validation-side work.

No universal functional form is assumed. AI is therefore permitted on both sides of the system. The theory does not assume that AI only generates candidates.

8.3 Candidate-validation backlog

Let B_t be unresolved validation workload carried into period t , and $\delta_B \in (0, 1]$ the fraction of unresolved workload that becomes obsolete, is abandoned, or otherwise leaves the active queue per period.

The backlog evolves as

$$B_{t+1} = \max \{0, (1 - \delta_B)B_t + \Lambda_t - \mu_t\} \quad (16)$$

Status: New Derivation 1 — Candidate-Validation Backlog.

Equation (16) is the mathematical core of the theory.

8.4 Steady state

Assume constant Λ , μ , and $\delta_B > 0$. In the interior positive-backlog regime:

$$B^* = (1 - \delta_B)B^* + \Lambda - \mu.$$

Therefore:

$$B^* = \frac{\Lambda - \mu}{\delta_B} \quad \text{for } \Lambda > \mu. \quad (17)$$

If $\Lambda \leq \mu$, the non-negativity condition yields $B^* = 0$.

Proposition 1 — Validation-Backlog Equilibrium. Under constant flows and positive exit rate, the active validation backlog is zero when capacity weakly exceeds effective candidate workload and positive in proportion to the candidate-capacity gap when workload exceeds capacity.

8.5 AI-induced epistemic scarcity shift

Separate AI used for generation from AI used for validation. Let A_g increase generation-side capacity and A_v increase validation-side capacity:

$$\Lambda = \Lambda(A_g, \dots), \quad \mu = \mu(A_v, E^C, E^I, W, D, R).$$

In the positive-backlog regime:

$$\frac{\partial B^*}{\partial A_g} = \frac{\partial \Lambda / \partial A_g}{\delta_B} > 0 \quad (18)$$

when generation-side AI raises effective candidate workload, and

$$\frac{\partial B^*}{\partial A_v} = -\frac{\partial \mu / \partial A_v}{\delta_B} < 0 \quad (19)$$

when validation-side AI expands validation capacity.

Now let z represent a general AI capability shock that affects both sides. Then:

$$\frac{\partial B^*}{\partial z} = \frac{\partial \Lambda / \partial z - \partial \mu / \partial z}{\delta_B} \quad (20)$$

This gives the central condition:

$$\frac{\partial \Lambda}{\partial z} > \frac{\partial \mu}{\partial z} \iff \frac{\partial B^*}{\partial z} > 0 \quad (21)$$

Proposition 2 — AI-Induced Epistemic Scarcity Shift. In the positive-backlog regime, a general AI capability improvement increases unresolved validation workload if and only if its marginal effect on effective candidate workload exceeds its marginal effect on validation capacity.

This proposition is deliberately symmetric. It permits three empirical outcomes:

- **generation-dominant AI:** backlog rises;
- **balanced AI:** backlog is unchanged;
- **validation-dominant AI:** backlog falls.

The theory is therefore falsifiable and does not treat AI as inherently epistemically harmful.

8.6 Validated throughput and value

Let $\mathcal{P}_t \subseteq B_t \cup \mathcal{C}_t^{new}$ denote the candidates actually processed during period t , subject to workload capacity. Let $G(c) \in \{0, 1\}$ be the declared validation gate for candidate c , and let $v(c) \geq 0$ denote its realized epistemic or decision value conditional on passing the gate.

Define validated output value:

$$Y_{K,t} = \sum_{c \in \mathcal{P}_t} v(c) \mathbf{1}[G(c) = 1] \quad (22)$$

Status: New Definition 3.

Candidate abundance alone therefore does not imply proportional validated value:

$$|\mathcal{C}_t^{new}| \uparrow \not\Rightarrow Y_{K,t} \uparrow \text{ proportionally} \quad (23)$$

8.7 Bottleneck revaluation

Suppose candidate supply is already sufficient to saturate validation capacity. Locally, an additional unit of candidate-generation capacity need not increase processed validated output, while an additional unit of validation capacity can.

This is the economic core of the scarcity shift:

Once candidate generation ceases to be the binding constraint, the shadow value moves toward the resources that relax validation.

Let an institution maximize

$$\max_x U(Y_K) - C(x)$$

subject to

$$Y_K \leq \Phi(\mu, \mathcal{C}),$$

where Φ is the maximum validated value attainable from available candidates given validation capacity. The Lagrangian is

$$\mathcal{L} = U(Y_K) - C(x) + \lambda_V [\Phi(\mu, \mathcal{C}) - Y_K] \quad (24)$$

The multiplier λ_V is the shadow value of relaxing the validation constraint.

Proposition 3 — Bottleneck Revaluation. When the validation constraint binds and sufficiently valuable unprocessed candidates exist, the marginal social value of resources that expand μ is positive. A technology that expands candidate supply without proportionately expanding μ can therefore raise the shadow value of contributory expertise, interactional translation, world access, verification infrastructure, and reviewer attention even while it lowers the marginal cost of producing plausible answers.

This proposition concerns the constrained capability, not observed wages. Market power, ownership, institutions, and sector-specific labour substitution can prevent shadow value from appearing as higher pay.

9. Why Practitioners Become Epistemically Central

Practitioners should not be romanticized as infallible owners of reality. They can misremember, generalize from biased samples, defend organizational interests, and possess local knowledge that does not transfer. Their importance is narrower and stronger: practitioners often possess world contact that academic systems cannot reconstruct from literature alone.

They encounter repeated failure, tacit constraints, timing, logistics, customer behaviour, local regulation, equipment limitations, trust, informal workarounds, and consequences

that may never enter academic datasets.

In the Toledo knowledge representation, these contacts can contribute traces that later become structured and capacity-generating. They are not automatically validated knowledge, but they can contain distinctions that disappear when research begins only from an academic literature.

For practice-linked inquiry, the researcher's role is therefore not simply to produce knowledge and transfer it to a passive user. The task is to organize a **coupled epistemic system** in which different participants contribute non-equivalent resources.

The proposed cycle is

$$\begin{array}{l}
 \text{Live Problem} \rightarrow \text{Core Respondent / Practitioner} + \text{Researcher} \\
 + \text{Interactional Expert if present} + \text{AI Model(s)} \\
 \rightarrow \text{Question} \rightarrow \text{Candidate Set} \rightarrow \text{Discrimination} \\
 \rightarrow \text{World Test} \rightarrow \text{Feedback} \rightarrow \text{Revision.}
 \end{array} \tag{25}$$

Equation (25) is a research architecture, not a linear law. Projects can branch, cycle, terminate, or return to earlier stages.

10. Human Return: Productivity Is Not Capability

A Toledo Human Return representation is

$$R_H^{\text{return}} = \langle C, T, S, A \rangle \tag{26}$$

where the four components represent retained conceptual reconstruction, tool selection, skill execution, and alternative generation after AI assistance.

The relevant non-collapse is

$$\Delta \text{Performance}_{AI} > 0 \not\Rightarrow \Delta R_H^{\text{return}} > 0 \tag{27}$$

This distinction changes how universities should evaluate AI adoption. A researcher may produce a better manuscript, faster code, or more candidate hypotheses with AI while becoming less able to reconstruct the reasoning, select tools independently, detect errors, or generate alternatives after the AI is removed.

The policy target cannot therefore be maximum AI usage. It should be **justified assistance with preserved or expanded human capability**, especially when future independent judgment matters.

11. Early-Career Researchers as an Incentive Equilibrium

11.1 Strategic conformity is rational under current payoffs

Let an early-career researcher choose project j from a set of feasible projects. Each project carries expected private career payoff

$$U_{ij}^{career} = f(p_{pub}, p_{grant}, T_{completion}, S_{supervisor}, V_{signal}, R_{failure}, \dots).$$

A project with lower scientific upside but high publication probability and short completion time can dominate a high-upside exploratory project when employment and promotion depend on short-window visible outputs.

This is an equilibrium problem, not a moral defect in researchers.

11.2 AI can reinforce the equilibrium

If AI lowers the cost of already-counted outputs, the institutional baseline can adjust upward:

1. AI increases the feasible volume of drafts, analyses, and manuscripts;
2. departments and journals face more output;
3. expected productivity rises;
4. researchers adopt AI to meet the new baseline;
5. low-variance work becomes cheaper to reproduce;
6. validation, field access, and world-testing capacity do not necessarily grow;
7. risky projects remain career-dangerous.

The result can be **output inflation without expertise formation**.

11.3 The alternative equilibrium

The alternative is not an anti-publication regime. Publication remains a vital public record. The change is that publication loses its monopoly as the main institutional readout of research value.

For practice-linked research, institutional value should also be attached to:

- a consequential live problem;
- transparent epistemic registration;
- candidate provenance;
- independent discrimination;
- world-side testing where appropriate;
- correction after failure;
- transfer to novel cases;
- and retained human capability.

Changing these rewards changes the private return to expertise-forming research.

12. Two Research Tracks, Not One

The framework explicitly rejects

$$\text{Problem-First} = \text{Problem-Only}.$$

12.1 Track A — Live-Problem / Translational / Use-Inspired Research

Entry condition: a consequential unresolved difference with an identifiable world-side owner, affected group, or operational context.

Typical architecture:

$$\text{Live Problem} \rightarrow \text{Practice Coupling} \rightarrow \text{Inquiry} \rightarrow \text{World Test} \rightarrow \text{Revision}.$$

Evaluation should emphasize validity, world closure, correction, transfer, learning, and capability conversion.

12.2 Track B — Open Exploration / Basic Research

Entry condition: a scientifically legitimate open question that may have no identifiable user or current application.

This track protects:

- foundational theory;
- curiosity-driven research;
- long-horizon basic science;
- high-risk conceptual exploration;
- and uses that cannot be predicted ex ante.

The architecture is compatible with Stokes's insight that understanding and use are not a single binary. Research can be use-inspired and fundamental at the same time.

The reform must therefore avoid replacing **publish-or-perish** with **impact-or-perish**.

13. University Objective: A Constrained Portfolio, Not a Universal Score

A universal scalar research score would reproduce the measurement problem the paper criticizes. It would force heterogeneous epistemic values onto one scale and invite gaming.

Instead, let university policy π_U seek a Pareto-efficient portfolio over

$$\boxed{\{Y_K, \Delta R_H^{return}, W, T, N_v\}} \quad (28)$$

where:

- Y_K = validated knowledge value;
- ΔR_H^{return} = retained human capability;
- W = world closure where appropriate;
- T = transfer to novel cases or users;
- N_v = epistemically relevant novelty or diversity.

Subject to:

$$\boxed{\begin{array}{l} \text{Budget} \leq \bar{B}, \\ \text{Safety risk} \leq \bar{S}, \\ \text{Protocol burden} \leq \bar{U}, \\ \text{Provenance and legitimacy gates passed,} \\ \text{Open-exploration capacity protected.} \end{array}} \quad (29)$$

Publication, citation, commercialization, and policy uptake may all enter project-specific evaluation, but no single indicator is treated as a universal proxy for research value.

14. Three Simultaneous Reform Layers

The theory implies that isolated interventions will be unstable. If universities change but funders do not, old incentives return through grant criteria. If funders change but promotion systems do not, researchers still face career penalties. If AI access expands without validation infrastructure, candidate workload can simply grow faster.

14.1 Layer 1 — Researcher, Practitioner, and AI Collaboration

For practice-linked projects:

1. Start from a live problem, or explicitly declare an open-exploration route.
2. Register the Core Respondent / Experience-Based Expert.
3. Register the Interactional Expert or `None`.
4. Register every materially used AI model and its role.
5. Preserve raw practitioner/problem traces before theoretical rewriting.
6. Use AI to expand candidate space, translate literature, generate rivals, and expose assumptions.
7. Use human and independent routes to discriminate among candidates.
8. Test action-relevant claims against world constraints.
9. Record failed predictions and model revisions as legitimate scientific outputs.

10. Measure what capability remains with human participants after AI assistance.

14.2 Layer 2 — University Reform

Universities should build infrastructure that increases μ , not only infrastructure that increases Λ :

- **Live Problem Registries** through which firms, hospitals, schools, local governments, social enterprises, and communities can submit unresolved problems;
- **Practice Partnerships** that give early-career researchers protected time in operational settings;
- **Interactional Expertise Roles** recognized as intellectual labour;
- **Shared AI Infrastructure** so frontier-model access is not determined only by laboratory wealth;
- **Validation Infrastructure** including replication support, data stewardship, methods consultation, adversarial review, and protected reviewer time;
- **Two-Track Evaluation** separating open exploration from problem-linked research;
- **Portfolio Evaluation** rather than universal scalar ranking;
- **Longer Time Horizons** for high-variance work;
- **Correction Credit** for documented revision, negative results, and falsification.

14.3 Layer 3 — Government and Funding Reform

Public funders should allocate part of the research portfolio to **problem-expertise formation**, not merely project output.

Problem-linked grant applications should identify:

- the live problem;
- the Core Respondent / world-side problem owner;
- relevant contributory expertise;
- interactional expertise or **None**;
- AI systems to be used;
- candidate-generation route;
- validation route;
- world-test route;
- revision protocol;
- and expected capability transfer.

Practitioners should be eligible for funded intellectual roles rather than appearing only as beneficiaries or data providers.

Funding should preferentially expand scarce capacities that generative AI does not automatically make abundant: field access, longitudinal observation, equipment,

replication, translation, verification, independent review, and contributory expertise.

15. How the Theory Extends Haubrock et al.

Haubrock et al. structural diagnosis	Extension developed here
Short cycles favor tractable work	Model the career system as an incentive environment that shapes expertise-forming trajectories
Productivity metrics reward safe outputs	Show that AI makes some counted outputs cheaper, increasing the risk of metric-output inflation
Hierarchy and precarity suppress creativity	Treat dependence as restricting accessible problem and hypothesis spaces before explicit choice
Collaboration should be valued	Specify distinct project roles and human expertise types rather than treating collaboration as homogeneous
Institutional inequality concentrates talent	Add unequal access to AI, validation capacity, world testing, and interactional bridges
Creativity is an emergent property of supportive systems	Extend emergence from creativity to expertise formation and retained human capability
Stable pathways and decentralized evaluation are needed	Add candidate-validation economics, Core Epistemic Registration, two research tracks, and world-feedback closure
System reform is required	Specify coordinated reform across researcher-practitioner practice, universities, and public funding

The relationship is therefore cumulative rather than oppositional. Haubrock et al. identify a structural creativity problem. The present paper asks how generative AI changes the scarcity structure underneath that problem and why the relevant institutional target should expand from creativity protection to expertise formation.

16. Testable Predictions

A theory of research reform should state what would count against it.

H3 — Candidate Workload Inflation

Departments with rapid generative-AI adoption should show a larger increase in effective candidate-validation workload than in validated output unless validation capacity expands simultaneously.

Falsifier: validated throughput rises at the same or a faster rate without additional validation resources or improvements in validation-side AI.

H4 — Scarcity-Shift Condition

When an exogenous AI capability improvement produces

$$\partial\Lambda/\partial z > \partial\mu/\partial z,$$

validation backlog or validation-equivalent burden should increase.

Falsifier: repeated settings with generation-dominant AI shocks show no rise in backlog or burden despite fixed exit rates and validation resources.

H5 — Interactional Compression

AI-supported researchers entering adjacent domains should reach pre-specified interactional-competence thresholds faster than matched controls in domains where language and literature access are major barriers.

Falsifier: no improvement in blinded interactional-expertise assessments despite adequate AI use and training exposure.

H6 — Contributory Persistence

In practice-intensive domains, AI-only learning should produce smaller gains in contributory performance than AI combined with real practice exposure.

Falsifier: AI-only participants match practice-exposed participants on blinded, real-world contributory tasks across repeated domains.

H7 — Practitioner Coupling

For practice-linked research, projects with registered practitioner coupling and world-feedback closure should show higher rates of successful transfer, correction, or sustained use than matched publication-first projects.

Falsifier: no difference or worse performance after credible adjustment for selection into collaboration.

H8 — Human-Return Dissociation

Immediate AI-assisted research performance can improve without proportional delayed unaided capability.

Falsifier: assisted-performance gains consistently predict equal or larger delayed unaided gains across tasks, expertise levels, and removal tests.

H9 — Incentive Recomposition

Departments that credibly reward world closure, correction, transfer, and high-variance inquiry should increase the share of early-career projects undertaking practice-linked or exploratory research without reducing research integrity.

Falsifier: credible incentive reform leaves research portfolios unchanged or materially lowers integrity.

17. Empirical Research Programme

The theory is global; Thailand is proposed as an initial implementation and replication test bed rather than as the boundary of the theory.

17.1 Multi-university design

A stepped-wedge or cluster-randomized design could compare four conditions:

1. **Conventional Research Condition** — existing departmental practice.
2. **AI Assistance Only** — approved AI access and training without research-architecture change.
3. **Live Problem + Practitioner + AI** — mandatory live-problem entry, Core Respondent registration, AI disclosure, and planned world-feedback closure.
4. **Live Problem + Practitioner + Interactional Expert + AI** — Condition 3 plus a registered human interactional expert where a qualified person is genuinely available.

The fourth condition must not force artificial role-filling. **None** is a legitimate and informative state.

17.2 Measurement architecture

Primary outcomes should include:

- effective candidate-validation workload Λ ;
- validation capacity μ ;
- active backlog B ;
- time from problem registration to testable question;
- number and diversity of non-equivalent hypotheses;
- time to first world test;
- proportion of processed candidates passing a declared gate;
- number and quality of revisions after world feedback;
- transfer to novel settings;
- practitioner-rated usefulness;
- independent methodological quality;
- publication quality;
- delayed unaided Human Return;
- protocol burden;

- and cost per validated, transferred output.

Secondary outcomes should include:

- collaboration-network diversity;
- cross-disciplinary language competence;
- AI dependency;
- model monoculture;
- negative-result retention;
- publication delay;
- reviewer workload;
- research-career retention;
- and distribution of opportunities across institution types.

17.3 Measuring Λ and μ

Because candidate claims differ in validation cost, empirical work should estimate $w(c)$ using pre-registered workload classes or time-and-resource estimates rather than raw claim counts. Validation capacity should be measured in the same units using completed validation-equivalent workload per period, adjusted for quality and route independence.

The theory does not require perfect measurement. It requires that candidate workload and validation capacity be measured on a compatible declared scale.

17.4 Identification concerns

The empirical programme must address:

- selection into practitioner collaboration;
- differential AI literacy;
- discipline-specific validation cost;
- problem difficulty;
- prestige and network effects;
- access to equipment;
- publication lag;
- practitioner conflicts of interest;
- and self-selection of stronger researchers into harder problems.

Randomization, matched designs, pre-registration, blinded outcome assessment, longitudinal follow-up, and difference-in-differences designs should be used where feasible.

18. Rival Explanations and Adversarial Tests

A world-level theory should not only state supporting mechanisms. It should specify rival models that can generate similar observations.

Rival Model R1 — Pure Productivity Transition

AI increases both candidate generation and validation so strongly that any temporary backlog is transitional. Under R1,

$$\frac{\partial \mu}{\partial z} \geq \frac{\partial \Lambda}{\partial z}$$

in mature adoption.

Discriminating evidence: longitudinal estimates of generation and validation elasticities after organizations have adapted.

Rival Model R2 — Metric Congestion Without Epistemic Scarcity

Observed review delays and publication congestion may reflect journal capacity rather than a deeper validation bottleneck. Candidate abundance could still translate into knowledge through other routes.

Discriminating evidence: measure world-testing, replication, verification, and decision-review burden outside journals. If only submission queues grow, the broader scarcity-shift claim is weakened.

Rival Model R3 — Expertise Substitution

AI may substitute for significant components of interactional and contributory expertise, making world-grounded expertise less scarce rather than more scarce.

Discriminating evidence: blinded comparisons of AI-only, human-only, and coupled systems on contributory real-world tasks with consequences, not merely verbal assessments.

Rival Model R4 — Institutional Inertia Only

The main problem may be slow policy adaptation rather than a persistent economic bottleneck. Once promotion systems and funding rules update, candidate and validation capacities may rebalance without a special expertise framework.

Discriminating evidence: institutions that change incentives but do not invest specifically in validation or practice access. If they close the gap anyway, the expertise bottleneck is less central than proposed.

These rivals prevent the theory from explaining every possible outcome after the fact.

19. Risks and Failure Modes

19.1 Practitioner capture

A live-problem architecture can become contract research in which the practitioner controls the question, evidence, and acceptable answer.

Safeguards: agenda co-ownership, conflict-of-interest disclosure, independent validation, and publication freedom.

19.2 Credential reversal

Rejecting credentialism must not create the opposite rule that experience automatically confers technical authority.

Safeguard: experience is registered as a source of problem contact, while claims remain domain-specific and subject to validation.

19.3 AI as invisible interactional expert

AI can generate fluent disciplinary language, creating the illusion that a human team has achieved cross-domain understanding.

Safeguard: AI remains registered as AI. If no human interactional expert exists, the field remains `None`.

19.4 Impact-or-perish

Universities may overcorrect by demanding immediate usefulness from all research.

Safeguard: protected open-exploration and basic-research portfolios.

19.5 Validation bureaucracy

The framework itself can create documentation overhead.

A Toledo protocol-burden representation is

$$B^{use} = \langle Time, CognitiveLoad, VerificationCost, Interruption, LiteracyDemand \rangle \quad (30)$$

The design principle should therefore be **minimum sufficient auditability**, not maximal documentation.

19.6 AI inequality

Frontier models, proprietary data, compute, and validation infrastructure may concentrate in wealthy universities and firms.

Safeguard: shared public AI access, pooled validation infrastructure, and explicit distributional monitoring.

19.7 Model monoculture

Using multiple AI models does not guarantee independence if they share training data, architectures, benchmarks, or institutional priors.

Safeguard: model plurality should be treated as one form of diversification, not as proof of independent validation.

20. Global Policy Implications

The theory is not primarily a proposal for one national university system. It identifies a general institutional transition.

The twentieth-century research university was built under a scarcity regime in which access to literature, analytical labour, expert language, computation, and polished scientific output were comparatively expensive. Generative AI reduces several of these costs. The university's comparative advantage should therefore move toward the activities that remain difficult to commoditize: trusted access to the world, durable communities of expertise, high-quality validation, long-horizon risk pooling, open criticism, and protection of inquiry that cannot justify itself through immediate market demand.

This has three global implications.

First, **AI policy for universities should be evaluated by task allocation, not adoption rate.** The important distinction is not AI versus no AI, but AI used for generation, AI used for validation, and the human capabilities retained after both.

Second, **research inequality may increasingly be inequality in validation capacity.** Shared access to a language model is not equivalent to shared access to equipment, longitudinal data, practice settings, independent reviewers, or contributory experts.

Third, **the public value of universities may rise, not fall, if they reorganize around trusted validation and expertise formation.** AI can commoditize some academic outputs while increasing the importance of institutions that can establish when an output deserves epistemic weight.

21. Limitations

This manuscript has eight important limitations.

First, it is a theory paper. The new equations identify relations to test; they are not empirically estimated results.

Second, the backlog model is intentionally minimal. Real validation queues contain dependencies among claims, variable urgency, strategic abandonment, heterogeneous failure costs, and endogenous prioritization.

Third, $w(c)$ is a measurement construct. Different empirical domains will require different workload units and calibration.

Fourth, Collins and Evans's expertise categories are ideal types rather than naturally discrete numerical states. Equation (7) is a research readout, not an ontological claim.

Fifth, the Core Respondent / Experience-Based Expert is a project role, not a fourth mutually exclusive SEE expertise state. The same individual can also hold interactional or contributory expertise if separately justified.

Sixth, AI can increase validation capacity. The theory explicitly permits $\partial\mu/\partial z > 0$; the central scarcity-shift claim is conditional on the relative elasticities of generation and validation.

Seventh, not every important form of research has a world-side user or practitioner. Open exploration is therefore preserved as a separate track.

Eighth, the theory does not yet provide a validated scalar measure of expertise, creativity, or research value. This omission is intentional. Premature scalarization would recreate the metric pathologies the framework seeks to avoid.

22. Discussion

The economics of expertise begins from a simple observation: abundance at one layer of a knowledge system can create scarcity at another.

Generative AI makes academically recognizable form cheap. It can produce explanations, summaries, code, candidate hypotheses, alternative framings, and polished prose at speeds that were previously impossible for individual researchers. That is a genuine expansion of productive capacity.

But a scientific system is not only a text-generation system. It must decide which distinctions matter, which claims are non-equivalent, which evidence is decisive, which source actually supports which inference, when a model has failed, and whether a result survives contact with the world.

When candidate production was expensive, production itself could function as a rough signal of capability. As generation becomes cheaper, that signal weakens. The scarce asset shifts toward **grounding**.

Grounding is distributed. Researchers possess theoretical and methodological structures. Practitioners possess repeated contact with operational constraints. Interactional experts can translate between communities. Contributory experts can perform and judge domain

practice. AI can lower the cost of search, translation, candidate expansion, and some forms of verification. None of these roles is sufficient alone.

This is why the three-part Core Epistemic Registration is more than an AI disclosure statement. It asks a paper to expose the structure through which problem contact, cross-domain translation, and machine mediation entered the inquiry. The registration does not certify truth. It makes missing roles and mediation visible.

The larger intervention is institutional. If universities reward only grants, publications, and citation-visible outputs, generative AI can make the existing regime more productive without making it more epistemically valuable. If universities instead invest in validation, correction, world contact, transfer, retained human capability, and protected open exploration, AI can become an accelerator of expertise formation rather than merely an accelerator of academic production.

The theory therefore reframes the early-career problem. The question is not only how to keep young researchers creative. It is how to build a system in which the career-rational strategy is compatible with becoming the kind of researcher who can enter difficult problems, cross expertise boundaries, survive being wrong, revise under evidence, and return to the world with greater capability.

23. Conclusion

Haubrock et al. (2026) are right to treat scientific creativity as an emergent property of academic systems rather than a stable trait of individual researchers. The next question is what those systems are optimized to produce in an era of generative AI.

This paper argues that four distinctions become load-bearing:

$$K_{\text{like}}^{AI} \neq K_{\text{validated}},$$

$$\text{knowledge} \neq \text{expertise},$$

$$\text{interactional expertise} \neq \text{contributory expertise},$$

and

$$\text{AI-assisted performance} \neq \text{retained human capability}.$$

From these distinctions follows the paper's central economic result. In a positive-backlog regime,

$$\frac{\partial B^*}{\partial z} = \frac{\partial \Lambda / \partial z - \partial \mu / \partial z}{\delta_B}.$$

AI increases epistemic scarcity when it increases effective candidate workload faster than it increases validation capacity. It reduces scarcity when the opposite is true. The theory therefore replaces a general claim that "AI creates too much information" with a testable condition about relative capacity growth.

The institutional implication is equally specific. Practice-linked research should move from

Academic → Topic → Grant → Study → Paper → User

toward

Live Problem → Practitioner + Researcher → AI-assisted Inquiry → World Test → Feedback →

Basic and open-ended research should remain protected outside this applied architecture.

The policy challenge is therefore not simply to motivate early-career researchers or teach them to use AI. It is to reorganize the economics of research so that the scarce capacities of the AI era—validation, judgment, world contact, correction, interactional translation, contributory expertise, and durable human capability—receive institutional support commensurate with their value.

If that reorganization occurs, generative AI can lower the cost of crossing intellectual boundaries without collapsing fluency into knowledge or assistance into expertise. If it does not, universities may become increasingly efficient at producing the safe, legible, strategically conforming outputs that the current reform debate is trying to escape.

Declarations

Author Contributions

Yaoharee Lahtee: conceptualization, theory development, framework design, mathematical derivation, literature integration, policy interpretation, manuscript revision, and final responsibility for all claims.

AI Assistance Disclosure

GPT-5.6 Sol was used for conceptual synthesis, equation-provenance audit, rival-model generation, mathematical restructuring, literature organization, adversarial review, and drafting assistance. Claude Fable 5.1 (via Claude Code) was used for the independent glosa review, the registration of this paper's equations in the Toledo equation library, PDF typesetting, and the Zenodo deposit. AI assistance does not constitute authorship. The human author retains responsibility for source selection, interpretation, mathematical proposals, claims, and the submitted text.

Interactional Expertise Disclosure

No human Interactional Expert was registered for this manuscript. This absence is disclosed rather than imputed to the AI system.

Funding

No external funding is declared for this theoretical manuscript.

Competing Interests

The author declares no competing interest specific to the argument of this manuscript. Any journal-specific institutional or commercial disclosure should be added at submission if required.

Ethics

This manuscript is conceptual and reports no new human-participant or animal-participant data.

Data Availability

No new empirical dataset was generated.

Code Availability

No empirical estimation code accompanies v1.0. The Toledo equation registry used for equation provenance is publicly available through its cited repository and Zenodo record.

References

- Aghion, P., Dewatripont, M., & Stein, J. C. (2008). Academic freedom, private-sector focus, and the process of innovation. *The RAND Journal of Economics*, 39(3), 617–635. <https://doi.org/10.1111/j.1756-2171.2008.00031.x>
- Arrow, K. J. (1962). Economic welfare and the allocation of resources for invention. In *The Rate and Direction of Inventive Activity: Economic and Social Factors* (pp. 609–626). Princeton University Press / National Bureau of Economic Research. <https://doi.org/10.1515/9781400879762-024>
- Azoulay, P., Graff Zivin, J. S., & Manso, G. (2011). Incentives and creativity: Evidence from the academic life sciences. *The RAND Journal of Economics*, 42(3), 527–554. <https://doi.org/10.1111/j.1756-2171.2011.00140.x>
- Bikard, M., Vakili, K., & Teodoridis, F. (2019). When collaboration bridges institutions: The impact of university–industry collaboration on academic productivity. *Organization Science*, 30(2), 426–445. <https://doi.org/10.1287/orsc.2018.1235>
- Brynjolfsson, E., Li, D., & Raymond, L. R. (2025). Generative AI at work. *The Quarterly Journal of Economics*, 140(2), 889–942. <https://doi.org/10.1093/qje/qjae044>

- Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1), 128–152. <https://doi.org/10.2307/2393553>
- Collins, H. M., & Evans, R. (2002). The third wave of science studies: Studies of expertise and experience. *Social Studies of Science*, 32(2), 235–296. <https://doi.org/10.1177/0306312702032002003>
- Collins, H., & Evans, R. (2007). *Rethinking Expertise*. University of Chicago Press. <https://doi.org/10.7208/chicago/9780226113623.001.0001>
- Dasgupta, P., & David, P. A. (1994). Toward a new economics of science. *Research Policy*, 23(5), 487–521. [https://doi.org/10.1016/0048-7333\(94\)01002-1](https://doi.org/10.1016/0048-7333(94)01002-1)
- Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28), eadn5290. <https://doi.org/10.1126/sciadv.adn5290>
- Edwards, M. A., & Roy, S. (2017). Academic research in the 21st century: Maintaining scientific integrity in a climate of perverse incentives and hypercompetition. *Environmental Engineering Science*, 34(1), 51–61. <https://doi.org/10.1089/ees.2016.0223>
- Etzkowitz, H., & Leydesdorff, L. (2000). The dynamics of innovation: From National Systems and “Mode 2” to a Triple Helix of university–industry–government relations. *Research Policy*, 29(2), 109–123. [https://doi.org/10.1016/S0048-7333\(99\)00055-4](https://doi.org/10.1016/S0048-7333(99)00055-4)
- Haubrock, P. J., Kouba, A., Soto, I., et al. (2026). How contemporary academic structures constrain scientific creativity and hold back early-career researchers. *Nature Human Behaviour*. <https://doi.org/10.1038/s41562-026-02575-5>
- Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., & Rafols, I. (2015). Bibliometrics: The Leiden Manifesto for research metrics. *Nature*, 520, 429–431. <https://doi.org/10.1038/520429a>
- Jones, B. F. (2009). The burden of knowledge and the “death of the Renaissance man”: Is innovation getting harder? *The Review of Economic Studies*, 76(1), 283–317. <https://doi.org/10.1111/j.1467-937X.2008.00531.x>
- Lahtee, Y. (2026a). *Toledo — Equation Library of the Human–AI Readout Programme* (Version 1.4.0). Open Civil Science Initiative. Zenodo concept DOI: 10.5281/zenodo.22537318; Version DOI: 10.5281/zenodo.22637913. Repository: <https://github.com/morrocwi/toledo>
- Lahtee, Y. (2026b). *glosa — Rigour Without Infrastructure: A Standalone Scholar Methodology for Human–AI Knowledge Co-Production* (Version 0.4.1). Open Civil Science Initiative. <https://doi.org/10.5281/zenodo.22340255>
- Manso, G. (2011). Motivating innovation. *The Journal of Finance*, 66(5), 1823–1860. <https://doi.org/10.1111/j.1540-6261.2011.01688.x>

- Nelson, R. R. (1959). The simple economics of basic scientific research. *Journal of Political Economy*, 67(3), 297–306. <https://doi.org/10.1086/258177>
- Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654), 187–192. <https://doi.org/10.1126/science.adh2586>
- Park, M., Leahey, E., & Funk, R. J. (2023). Papers and patents are becoming less disruptive over time. *Nature*, 613, 138–144. <https://doi.org/10.1038/s41586-022-05543-x>
- Polanyi, M. (1966). *The Tacit Dimension*. Doubleday. Reissued by University of Chicago Press. No DOI; publisher edition as listed (Doubleday, 1966), ISBN 0-385-06988-X.
- Stephan, P. (2012). *How Economics Shapes Science*. Harvard University Press. <https://doi.org/10.4159/harvard.9780674062757>
- Stokes, D. E. (1997). *Pasteur's Quadrant: Basic Science and Technological Innovation*. Brookings Institution Press. No DOI; publisher edition as listed (Brookings Institution Press, 1997), ISBN 0-8157-8177-6.
- Wang, J., Veugelers, R., & Stephan, P. (2017). Bias against novelty in science: A cautionary tale for users of bibliometric indicators. *Research Policy*, 46(8), 1416–1436. <https://doi.org/10.1016/j.respol.2017.06.006>

Appendix A. Equation and Claim Status Ledger

Eq.	Object	Status in this manuscript	Provenance
(1)	Economics-domain commuting requirement	Domain instantiation	Toledo domain-weld requirement
(2)	$K_{like}^{AI} \neq K_{validated}$	Existing non-collapse	Toledo CAN-062 lineage
(3)	AI-assisted performance $\neq$ retained human capability	Existing non-collapse / interpretation	Toledo Human Return lineage
(4)	Credential $\neq$ expertise	Methodological non-collapse	Collins–Evans + present framework; Toledo weld/H.22.v1
(5)	Project role $\neq$ expertise type $\neq$ AI	New methodological non-collapse	Present framework; Toledo weld/H.23.v1

Eq.	Object	Status in this manuscript	Provenance
(6)	$K_S^A = (Tr_A, Str_A, Cap_A)$	Existing knowledge architecture	Toledo knowledge-admission / knowledge-triple lineage (historical form of <code>weld/E.03.v1</code> ; see footnote in Section 5)
(7)	$\chi_i(Q, D, t) \in \{N, I, C\}$	New Definition	Formal readout of Collins–Evans ideal types; Toledo <code>weld/H.24.v1</code>
(8)	Expertise transition probability	Open Proposition	Present framework; Toledo <code>weld/H.25.v1</code>
(9)	Interactional compression	Open Hypothesis	Present framework; Toledo <code>weld/H.26.v1</code>
(10)	Contributory persistence	Open Hypothesis	Present framework; Toledo <code>weld/H.27.v1</code>
(11)–(13)	Core Epistemic Registration	New Definitions	Present framework; Toledo <code>weld/H.28.v1</code> (registration object), <code>weld/H.30.v1</code> – <code>weld/H.34.v1</code> (Core Epistemic Structure: experience-holder, AI model set, empty interactional-expert slot, non-collapse, experience decomposition)
(14)	Effective candidate workload	New Definition	Present framework; Toledo <code>weld/W.03.v1</code>
(15)	Validation capacity	New Definition	Present framework; Toledo <code>weld/W.04.v1</code>
(16)	Candidate-validation backlog	New Derivation	Present framework; Toledo <code>weld/W.05.v1</code>
(17)	Positive-backlog steady state	Derived from (16)	Present framework; Toledo <code>weld/W.06.v1</code>
(18)–(21)	AI comparative statics / scarcity-shift condition	New Derivation / Proposition	Present framework; Toledo <code>weld/W.07.v1</code>
(22)–(23)	Validated output value and non-equivalence	New Definition / non-collapse consequence	Present framework; Toledo <code>weld/W.08.v1</code>
(24)	Shadow-value Lagrangian	Standard constrained-optimization application	Present framework; Toledo <code>weld/W.09.v1</code>
(25)	Live-problem research architecture	New methodological architecture	Present framework + Toledo world-feedback closure; Toledo <code>weld/H.29.v1</code>
(26)	Human Return $\langle C, T, S, A \rangle$	Existing definition	Toledo Human Return lineage

Eq.	Object	Status in this manuscript	Provenance
(27)	Assisted performance does not imply Human Return	Existing non-collapse	Toledo
(28)– (29)	University multiobjective portfolio	New policy definition	Present framework; Toledo <code>weld/W.10.v1</code>
(30)	Protocol burden tuple	Existing definition	Toledo

Appendix B. Proof Sketches

B.1 Proposition 1 — Validation-Backlog Equilibrium

In the positive-backlog regime the max operator in Equation (16) is inactive. At steady state:

$$B^* = (1 - \delta_B)B^* + \Lambda - \mu.$$

Rearranging gives

$$\delta_B B^* = \Lambda - \mu,$$

hence

$$B^* = \frac{\Lambda - \mu}{\delta_B}.$$

If $\Lambda \leq \mu$, the unconstrained expression is non-positive and the non-negativity condition sets $B^* = 0$.

B.2 Proposition 2 — AI-Induced Epistemic Scarcity Shift

In the interior positive-backlog regime:

$$B^*(z) = \frac{\Lambda(z) - \mu(z)}{\delta_B}.$$

Holding δ_B constant:

$$\frac{\partial B^*}{\partial z} = \frac{\Lambda_z - \mu_z}{\delta_B}.$$

Since $\delta_B > 0$, the sign of $\partial B^*/\partial z$ equals the sign of $\Lambda_z - \mu_z$. Therefore backlog rises if and only if the AI shock raises effective candidate workload faster than validation capacity.

B.3 Proposition 3 — Bottleneck Revaluation

When candidate supply is sufficient and the validation constraint binds, feasible validated value is locally restricted by $\Phi(\mu, \mathcal{C})$. Under the standard regularity condition $\partial\Phi/\partial\mu > 0$, the Lagrange multiplier on the validation constraint is positive at an interior optimum with positive marginal utility of validated output. The shadow value of a resource that expands μ is therefore proportional to $\lambda_V \partial\Phi/\partial\mu$. This result does not imply any particular market wage because prices depend on ownership, bargaining, labour institutions, and substitutability.

Appendix C. Core Epistemic Registration Template

Core Epistemic Registration

Core Respondent / Experience-Based Expert:

[Name or anonymised role]

Basis of direct experience:

[Relationship to the problem, duration, responsibility, repeated exposure]

Interactional Expert:

[Name / role / basis of interactional expertise]

OR

None

AI Model(s) Used:

[Model name and version] – [role]

[Model name and version] – [role]

[Additional models if applicable]

Registration date:

[YYYY-MM-DD]

Changes after registration:

[Append-only log]

Appendix D. Applied-Research Project Card

```

project_id: ""
track: "live_problem" # or open_exploration

```

```

live_problem:
  statement: ""
  world_owner_or_affected_group: ""
  unresolved_difference: ""

```

```

core_epistemic_registration:
  experience_based_expert:
    name_or_role: ""
    basis: ""
  interactional_expert:
    status: "present" # or none
    name_or_role: ""
    basis: ""
  ai_models:
    - model: ""
      version: ""
      role: ""

```

```

candidate_workload:
  candidate_classes: []
  workload_scale: ""
  human_candidates: []
  ai_candidates: []
  rival_generation_method: ""

```

```

validation:
  capacity_unit: ""
  decisive_evidence: ""
  independent_route: ""
  world_test: ""
  falsifier: ""

```

```

human_return:
  concept_reconstruction: ""
  tool_selection: ""
  skill_execution: ""
  alternative_generation: ""

```

```

revision:
  failed_prediction_or_mismatch: ""
  structural_revision: ""
  next_problem: ""

```